Miscellaneous Algebraic Knowledge

Explain: How we set our tests and select our team, purpose of training, may talk about University stuff, no need to be restricted by syllabus.

Important General Things:

Idea (Concepts, definitions, approaches)
Scope (Depth, Range, complexity)
Machinery (Technique, theory, story)
Maturity (Read Intensively, Journal, chat)
Appreciation
Novel (Read like a novel)
No Mystery. ***Try to understand.***

Algebra, other areas. ***Geometry***, ***Analysis, Topology***.

Coordination, Quantification (Shafarevich)
Structure
Representation
Character

Build Up Number Systems, Integer, Rational, Irrational, Real. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.

Vectors, matrices (operations), vector products (dot and cross), multiplication, linear systems, determinants, Cramer's rule, vector spaces, basis. $\mathbb{R}^2, \mathbb{R}^3, \mathbb{R}^n, i, j, k, \mathbb{C}^2$. ways to find determinant, inverse of 2x2 matrices

Miscellaneous Algebraic Systems, Group (Lagrange), permutation groups, symmetry group of cube, Ring (CRT homomorphism, isomorphism, domain), Field, Vector (Space), Matrix, Finite Field, Coding Theory, Polynomials. Wedderburn.

TAKE THREE HOURS TO FINISH THE ABOVE.

Complex, Quaternions.

Complex Numbers, representation, Cartesian, polar, exponential, application (geometry and formula, most important, norm and geometric properties related to products of complex numbers), extension to 3-D, Hamilton story, Introduction to quarterions, unit quarterions, rotation, isomorphism. Illustration of how to define e^{iy} by expanding using Taylor series or by setting $e^{iy} = \cos y + i \sin y.$ Complex numbers isomorphic to real 2×2 matrices of the form $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$

16^{th}, October, 1843, mathematical vandalism, Brougham Bridge, Dublin, by W. R. Hamilton.

$$i^2 = j^2 = k^2 = ijk = -1.$$

Motivation

Try to multiply elements of the form $a+bi+cj,$ with $i^2 = j^2 = -1,$ and i, j independent norm 1 vectors.

Ideas:

(i) Assume $ij = ji,$ then

$$\begin{aligned} -1 &= i^2 = j^2 \Rightarrow 0 = i^2 - j^2 = i^2 - ij + ji - j^2 \\ &= i(i-j) + j(i-j) = (i+j)(i-j) \Rightarrow i = \pm j, \end{aligned}$$

contradiction. Hence $ij \neq ji.$

(ii) Assume $a+bi+cj$ a norm 1 vector, then

$$\begin{aligned} 1 &= a^2+b^2+c^2 = (a+bi+cj)(a-bi-cj) \\ &= a^2+b^2+c^2 - bc(ij+ji), \end{aligned}$$

hence should have $ij = -ji.$

(iii) Now what is ij? Try

$ij = a+bi+cj,$ get

$$\begin{aligned} -i &= ij^2 = (a+bi+cj)j \\ &= aj + bij - c = aj + b(a+bi+cj) - c \\ &= aj + ba + b^2 i + bcj - c \\ &= (c+ba) + b^2 i + (a+bc)j. \end{aligned}$$

Hence $b^2 = -1,$ a contradiction!

Example (***Exercise***). Using the definition of Hamilton, get $ij = k, kj = i, ji = k$. Also $ji = -k, jk = -i, ik = -j$.

Representations

A quaternion $a + bi + cj + dk$, if the situation is clear, is also usually written as (a,b,c,d).

Theorem. Sending a quaternion (a,b,c,d) to the 2×2 complex matrix $\begin{pmatrix} a+bi & -c-di \\ c-di & a-bi \end{pmatrix}$ is an $\mathbb{R}-$algebra isomorphism.

Likewise the real quaternions form an algebra which is isomorphic to the 4×4 real matrices:

$$\begin{pmatrix} a & -b & -c & -d \\ b & a & -d & c \\ c & d & a & -b \\ d & c & b & a \end{pmatrix}.$$

Conjugate, norm of a quaternion, finding inverse.

The *conjugate* of $q = a + bi + cj + dk$ is $\overline{q} = a - bi - cj - dk$. Its *norm* is $|q| = \sqrt{a^2 + b^2 + c^2 + d^2}$. Check that $q\overline{q} = |q|^2$. Note the inverse of q is $\dfrac{\overline{q}}{|q|^2}$.

Exercise. $\overline{q_1 q_2} = \overline{q_2}\,\overline{q_1}$.

Another form of writing a quaternion and a product of two of them.

Another way of writing a quaternion $a + bi + cj + dk$ as (a,u), where a is the same and u stands for the vector $bi + cj + dk$ (3-vector). Adding is the same component-wise, and product

$$(a_1,u_1)(a_2,u_2)=(a_1a_2-u_1\cdot u_2, a_1u_2+a_2u_1+u\times v).$$

Note the appearances of the dot and cross product.

Exercise. Check consistent with the original definition.

Unit quaternions and theorem about rotation. A quaternion $a+bi+cj+dk$ is a *unit quaternion* if $|q|=\sqrt{a^2+b^2+c^2+d^2}=1.$

Theorem: For any unit quaternion $q=\cos\frac{\theta}{2}+\sin\frac{\theta}{2}u,$ and for any $v\in\mathbb{R}^3,$ the action of the operator $L_q(v)=qvq^{-1}$ on v is equivalent to the rotation of v through an angle θ about u as the axis of rotation. Here v is regarded as the quaternion as $(0,v)$.

Exercise. (i) With the notation above, check $L_q(u)=u.$ (ii) What is L_{-q}? (iii) What is $L_{\cos\frac{\theta}{2}-\sin\frac{\theta}{2}u}$?

Euler Rotation Formula: Composition of rotations (with the centre always fixed) is a rotation.

A unit quaternion and its variations.

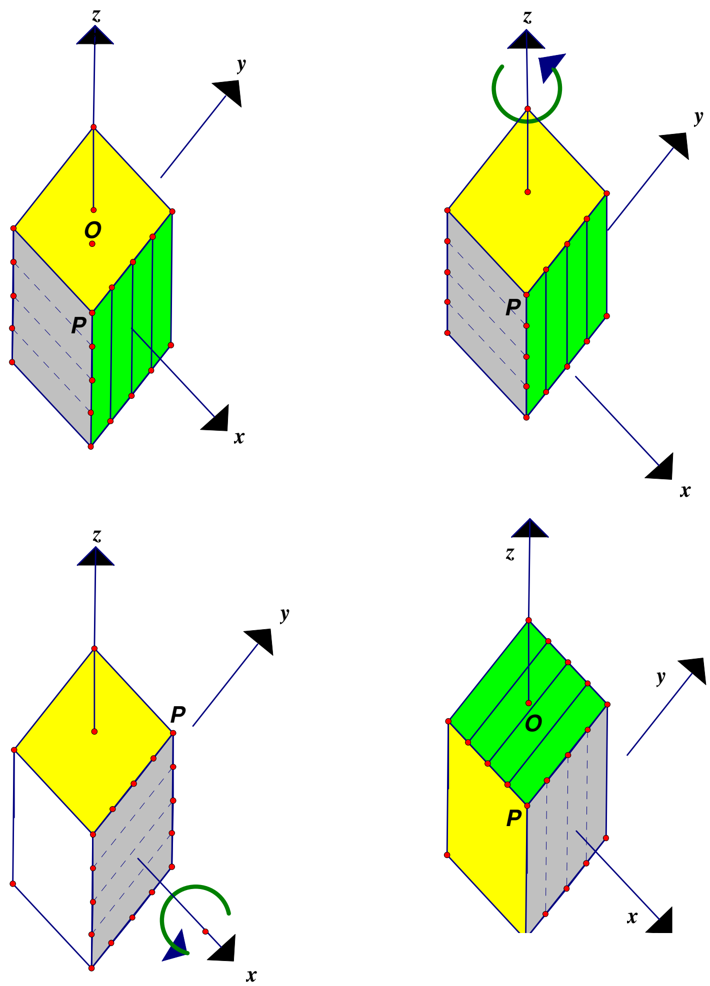
Example. First turn 90° about the z-axis, then 90° about the x-axis. What is the resulting rotation?

Turning 90° about the z-axis corresponds to the unit quaternion $q_1=\cos 45^{\circ}+\sin 45^{\circ}(0i+0j+\frac{1}{\sqrt{2}}k).$ Turning about the x-axis corresponds to the unit quaternion $q_2=\cos 45^{\circ}+\sin 45^{\circ}(\frac{1}{\sqrt{2}}i+0j+0k).$ Multiplying

$$q_2q_1=\frac{1}{2}+\frac{1}{2}i-\frac{1}{2}j+\frac{1}{2}k=\frac{1}{2}+\frac{\sqrt{3}}{2}(\frac{1}{\sqrt{3}}i-\frac{1}{\sqrt{3}}j+\frac{1}{\sqrt{3}}k)$$
$$=\cos 60^{\circ}+\sin 60^{\circ}(\frac{1}{\sqrt{3}}i-\frac{1}{\sqrt{3}}j+\frac{1}{\sqrt{3}}k).$$

Hence it is a rotation corresponds to turning about the axis $\frac{1}{\sqrt{3}}i-\frac{1}{\sqrt{3}}j+\frac{1}{\sqrt{3}}k \quad 120^{\circ}$.

The pictures illustrate the rotations.



(Note O is in the centre of the cube.)

Example. (i) Determine the image of the point $(1,-1,2)$ under the rotation of an angle 60° about an axis in the *yz*-plane that is inclined at an angle of 60° to the positive *y*-axis. (ii) Now consider the rotation of 60° about an axis in the *xz*-plane inclined at an angle to the positive *x*-axis. Determine the composition of the rotation in (i), the one in (i) acts first.

(i) The unit vector in the axis of rotation is $\cos 60^{\circ} j + \sin 60^{\circ} k$, hence the corresponding unit quaternion is

$$\cos 30^{\circ} + \sin 30^{\circ}(\cos 60^{\circ} j + \sin 60^{\circ} k) = \frac{\sqrt{3}}{2} + \frac{1}{2}(\frac{1}{2} j + \frac{\sqrt{3}}{2} j)$$

$$= \frac{1}{4}(2\sqrt{3} + j + \sqrt{3}k),$$

with its inverse $q^{-1} = \frac{1}{4}(2\sqrt{3} - j - \sqrt{3}k)$.

The corresponding image of $(1,-1,2)$ is therefore

$$q(1,-1,2)q^{-1} = q(0 + i - j + 2k)q^{-1} = \frac{1}{8}((10 + 4\sqrt{3})i + (1 + 2\sqrt{3})j + (14 - 3\sqrt{3})k).$$

(ii) The new unit vector of rotation is $\cos 60^{\circ} i + \sin 60^{\circ} k$, hence the new unit quaternion is

$$\cos 30^{\circ} + \sin 30^{\circ}(\cos 60^{\circ} i + \sin 60^{\circ} k) = \frac{\sqrt{3}}{2} + \frac{1}{2}(\frac{1}{2} i + \frac{\sqrt{3}}{2} k)$$

$$= \frac{1}{4}(2\sqrt{3} + i + \sqrt{3}k).$$

We then compute

$$\frac{1}{4}(2\sqrt{3} + i + \sqrt{3}k) \cdot \frac{1}{4}(2\sqrt{3} + j + \sqrt{3}k) = \frac{1}{16}(9 + \sqrt{3}i + \sqrt{3}j + 13k)$$

$$\frac{9}{16} + \frac{1}{16} \cdot 5\sqrt{7}(\frac{\sqrt{3}i + \sqrt{3}j + 13k}{5\sqrt{7}}) = \frac{9}{16} + \frac{5\sqrt{7}}{16}(\frac{\sqrt{21}i + \sqrt{21}j + 13\sqrt{7}k}{35}).$$

The corresponding rotation angle is $\cos^{-1}\frac{9}{16} \approx 55.77^{0}$.

Topology The fundamental group of $SU(2)$ is trivial, fundamental group of $SO(3)$ is $\mathbb{Z}_2$.

TAKE THREE HOURS TO TALK ABOUT COMPLEX NUMBERS AND QUATERNIONS etc, HARD TO DISCUSS FUNDAMENTAL GROUPS.

Exercise

1. The subset $\{\pm 1, \pm i, \pm j, \pm k\}$ forms a group.
2. Quaternions with rational coefficients form a division ring.
3. The equation $x^2 = -1$ has infinitely many quaternion solutions.
4. Verify the set *I* of quaternions with coefficients either are integers or halves of odd integers is a subring.
5. (Cartan-Brauer-Hua) Let *D* be a division ring, *C* its centre and let *S* be a division subring of *D* which is stabilized by every map $x \to dxd^{-1}, d \neq 0$ in *D*. Show that either $S = D$ or $S \subseteq C$.
6. Find the centre of all real quaternions.

Simple and Important Results (Chinese Remainder Theorem, Lagrange' Theorem, Cyclotomic Fields, Minimal Polynomials, Primitives, QR, examples

References

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